

Activity 1: Computing the Even Numbers in Two Ways

CHALLENGE

Can the same thing be made using all the numbers once and also using every other number once?

Task A: Doubling each term in a sequence

Train a robot called `Doubler` that receives a number on a nest and gives a bird twice that number. Illustrated instructions for training this robot are the end of this handout. Test out `Doubler` by giving it a box like this:



Give the `In` bird a number and watch what appears on the `Out` nest. Give the `In` bird a different number. If `Doubler` is

working as it should then whatever number you give the `In` bird you should get twice that number on the `Out` nest.

Clear off the `Out` nest of all the numbers. Give the `In` bird 1, and then 2, 3, 4, and 5.

What numbers appear on the `Out` nest?

Let's give `Doubler` all the *natural numbers* (1, 2, 3, and so on forever). Since we can't type in all the natural numbers we'll need a robot to help us. We'll call him `Add 1` since that is what he'll do over and over again. You'll find instructions for how to build `Add 1` at the end of this handout.

A natural number is a whole number greater than zero.



To test out `Add 1` give it a box like this:



and watch what numbers end up on the `Nat Nums` nest.

Try connecting `Doubler` to the sequence of *natural numbers* made by the `Add 1` robot. You can do this by giving `Doubler` the box:



Since the `In` hole is empty the robot won't do anything yet. Give the `Add 1` robot this box:



To get these robots to work together take out a nest with an egg in it. Give it a nice name. Drop the bird in the `Numbers` hole and the nest in the `In` hole.

Did you notice that the first numbers coming out are even?

Will there ever be an odd number?

Explain

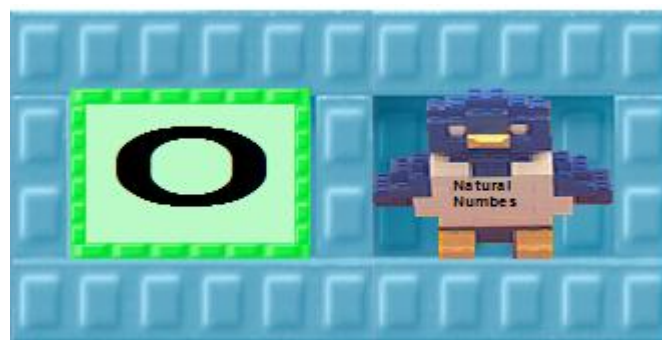
Remember to save your robots in your notebook!

Task B: Removing every other term in a sequence

Train a robot called `Split`. As numbers arrive on the `Input` nest `Split` gives them alternatively to bird A and B. Instructions for training `Split` are at the end of this handout.



Put the nest that receives the *natural numbers* produced by the `Add 1` robot in the hole labeled `Input` and give the box to `Split`. Give the `Add 1` robot this box:



What kind of numbers end up in A's nest?

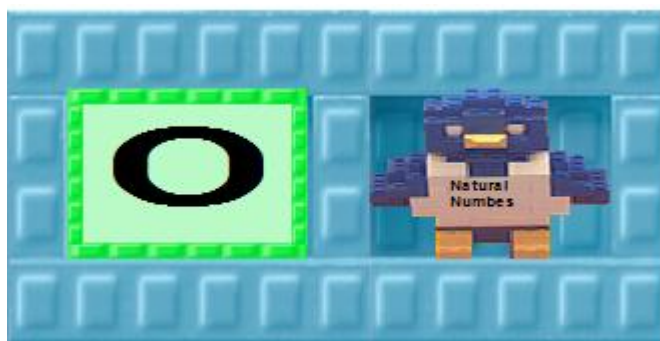
What kind of numbers end up in B's nest?

Remember to save your robots in your notebook

Task C: Thinking about the two ways of making all the even numbers

Copy the nest that will receive all the *natural numbers* produced by the Add 1 robot. Start up the Add 1 robot and you should see each copy of the nest receiving 1, 2, 3 and so on.

Start again and give one copy to your Doubler robot and the other to the Split robot. Now start up the Add 1 robot again with the box:



If all the robots ran forever would there be a difference between the numbers that are on B's nest and the output nest of Doubler?

If there is a difference, what is it?

If there is no difference, how can they be the same sequence if Doubler uses every *natural number* and Split puts only half of the numbers on B's nest?

Task D: Even numbers dancing with *natural numbers*

If you saw a room full of boys and girls and everyone was dancing with a partner of the opposite sex, could you figure out if there was the same number of boys as girls? Without counting!

How would you convince someone you are right?

Imagine that you are running the `Doubler` robot and that after the `Add 1` robots gives a bird a number it is copied and begins dancing with the output number of the `Doubler` robot. So 1 dances with 2, 2 with 4, 3 with 6, and so on.

If this continued forever would every even number be dancing with a *natural number*?

Would every *natural number* be dancing with an even number?

What does this tell you about how many even numbers there are?

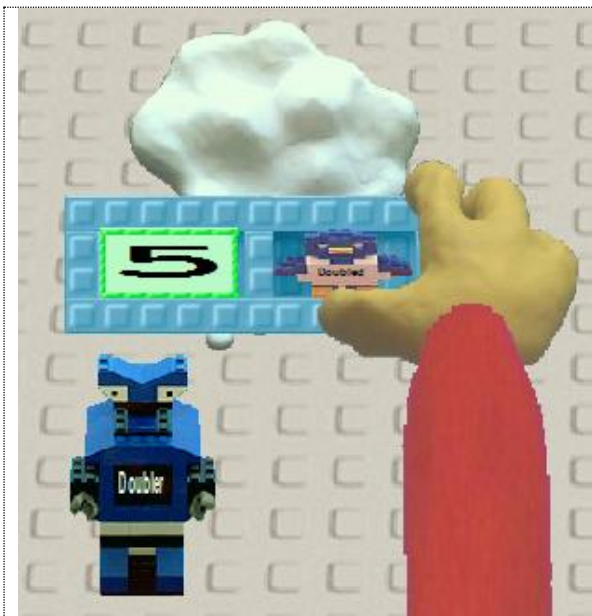
You may be confused at this point. Infinity confused some of the world's best mathematicians. Galileo Galilei wrote in 1638 about how one line of reasoning leads you to think the sequences have the same number of numbers while another line of reasoning leads you to think there are fewer even numbers.

What do you think?

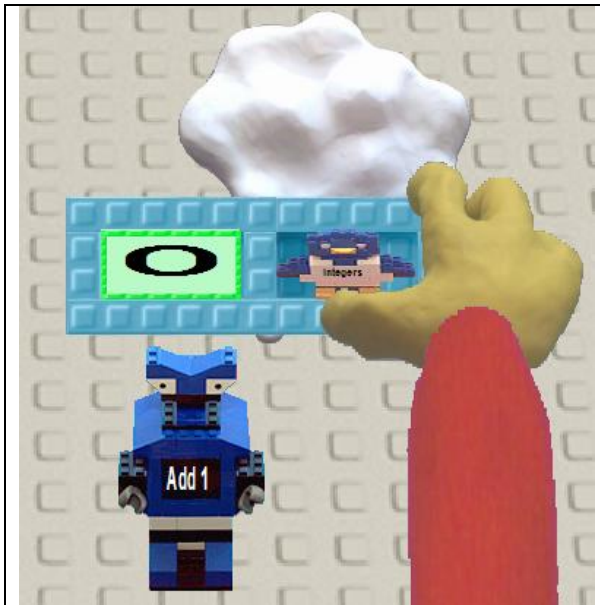
If you think they all pair up then how do you explain that the sequence of *natural numbers* contains all the even and all the odd numbers?

If you don't think they pair up, what is different from the room of boys and girls?

Activity 1, Task A: How to Train the Doubler Robot



Activity 1, Task A: How to Train the Add 1 Robot



Activity 1, Task B: How to Train the Split Robot

